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-- #d.u.m.b.a.s.s. (Database & Universal Memory Bank Across Split Services)
-- Master PostgreSQL Schema & Migrations (Version 1.0)
-- Target Host: Node Beta (NVIDIA Jetson Orin Nano / Headless Ubuntu Server)
-- Database Engine: PostgreSQL 16+ with pgvector & pgcrypto
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CREATE EXTENSION IF NOT EXISTS "uuid-ossf";
CREATE EXTENSION IF NOT EXISTS "pgcrypto";
CREATE EXTENSION IF NOT EXISTS "vector";

-- 1. Master Repositories & Subsystem Registry
CREATE TABLE IF NOT EXISTS repositories (
    repo_id VARCHAR(64) PRIMARY KEY, -- 'novaecopia', 'aesop-xi', 'novus-aexenti',
    'skills-and-capabilities'
    canonical_name VARCHAR(128) NOT NULL,
    root_path VARCHAR(256) NOT NULL,
    description TEXT,
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- 2. Universal Vector Memory Bank (Embeddings & Semantic Chunks)
CREATE TABLE IF NOT EXISTS vector_embeddings (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    repo_id VARCHAR(64) REFERENCES repositories(repo_id) ON DELETE CASCADE,
    file_path VARCHAR(512) NOT NULL,
    chunk_index INT NOT NULL,
    token_count INT NOT NULL,
    content TEXT NOT NULL,
    embedding vector(1536), -- Standard embedding dimension (Text-Embedding-3 / local BGE)
    metadata JSONB DEFAULT '{}':jsonb, -- Includes source headers, wikilinks, tags
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX IF NOT EXISTS idx_embeddings_repo ON vector_embeddings(repo_id);
CREATE INDEX IF NOT EXISTS idx_embeddings_metadata ON vector_embeddings USING
gin(metadata);

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-- HNSW index for high-speed approximate nearest neighbor search
CREATE INDEX IF NOT EXISTS idx_embeddings_hnsw ON vector_embeddings USING hnsw
(embedding vector_cosine_ops)
WITH (m = 16, ef_construction = 64);
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-- 3. Enterprise Audit & Execution Ledger (Immutable Telemetry)
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CREATE TABLE IF NOT EXISTS audit_ledger (
  log_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  session_id VARCHAR(128) NOT NULL,
  agent_role VARCHAR(64) NOT NULL, -- 'triage_0.8b', 'query_9b', 'executor_qwen',
  'claude_code'
  action_name VARCHAR(128) NOT NULL,
  input_payload JSONB,
  output_payload JSONB,
  verdict VARCHAR(32) NOT NULL DEFAULT 'PENDING', -- 'PASS', 'FAIL', 'QUARANTINED',
  'PENDING'
  reward_score NUMERIC(4, 2), -- e.g. +1.00 or -1.00
  latency_ms INT,
  node_origin VARCHAR(64) NOT NULL, -- 'node_alpha', 'node_beta', 'node_gamma'
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);
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CREATE INDEX IF NOT EXISTS idx_audit_session ON audit_ledger(session_id);
CREATE INDEX IF NOT EXISTS idx_audit_verdict ON audit_ledger(verdict);
CREATE INDEX IF NOT EXISTS idx_audit_created ON audit_ledger(created_at);
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-- 4. Quarantined & Failure Trace Register (Incognito Red Sandbox Integration)
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CREATE TABLE IF NOT EXISTS quarantine_traces (
  quarantine_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  log_id UUID REFERENCES audit_ledger(log_id) ON DELETE SET NULL,
  batch_hash VARCHAR(64) NOT NULL,
  source_path VARCHAR(512) NOT NULL,
  rejection_code VARCHAR(64) NOT NULL, -- 'SYNTAX_ERROR', 'HALLUCINATED_PATH',
  'SECURITY_VIOLATION'
  error_trace TEXT NOT NULL,
  is_resolved BOOLEAN DEFAULT FALSE,
  resolution_notes TEXT,
  quarantined_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);
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CREATE INDEX IF NOT EXISTS idx_quarantine_hash ON quarantine_traces(batch_hash);
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-- 5. Canonical Living Entity Knowledge Graph
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CREATE TABLE IF NOT EXISTS canonical_entities (
  entity_id VARCHAR(128) PRIMARY KEY,
  name VARCHAR(256) NOT NULL,
  entity_type VARCHAR(64) NOT NULL, -- 'system', 'subsystem', 'daemon', 'apk',
  'hardware_node'
  canonical_repo VARCHAR(64) REFERENCES repositories(repo_id) ON DELETE SET
  NULL,
  definition TEXT NOT NULL,
  properties JSONB DEFAULT '{}::jsonb,
  relations JSONB DEFAULT '[]::jsonb, -- Array of {target_id, relation_type}
  last_synced TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX IF NOT EXISTS idx_entities_type ON canonical_entities(entity_type);
CREATE INDEX IF NOT EXISTS idx_entities_props ON canonical_entities USING
gin(properties);

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